OPEN AGENT HACKATHON 2026 · TRACK T01 · NEMOTRON · TEAM ZEN

Probity: before the money moves, prove who is asking

Reads a vendor's bank-detail change, cross-examines every claim across sources that disagree, and decides: approve, call back first, or hold.



THE REQUEST: A SYNTHETIC LETTER, WATERMARK VISIBLE



THE DECISION, BEFORE THE PAYMENT RUN

SYNTHETIC LETTERS AND INTERNAL RECORDS · REAL USASPENDING AND SEC EDGAR DATA

WHO FEELS IT

Problem and users

Dana

AP CONTROLS LEAD · FICTIONAL FEDERAL INTEGRATOR

Owns the call-back policy and the payment-run release.

Priya

ASSISTANT CONTROLLER · SIGNS OFF THE RUN

Needs a hold list in dollars, not a mailbox.

Tomás

VENDOR-MASTER ANALYST · KEEPS THE RECORDS STRAIGHT

Applies changes, makes the call-back.

VENDOR MASTER — AP LEDGER — USASPENDING — SEC EDGAR — A PHONE

The moment: a bank-change request before a payment run. Policy says call back; practice calls the number on the letter.

$23.5 million

diverted after stolen contractor logins changed banking details on a government vendor payment system (2018).

USAO-NJ RELEASE 22-160 (29 APR 2022); GAO-26-108174

$3,046,598,558

lost to business email compromise in 2025.

FBI IC3 2025 INTERNET CRIME REPORT

91%

of organisations call back on a system-of-record number, not one in the email.

AFP 2025 PAYMENTS FRAUD AND CONTROL SURVEY (2024 DATA)

25–40 min

per request to find an independent number across 3–4 systems.

OUR ESTIMATE, NOT A MEASUREMENT

SYNTHETIC LETTERS AND INTERNAL RECORDS · REAL USASPENDING AND SEC EDGAR DATA

FOUR SOURCES, FOUR ANSWERS

The sources disagree

	CLAIM IN THE REQUEST	VENDOR MASTER	USASPENDING	SEC EDGAR
S1 · BOOZ ALLEN	Remit to 1 Commvault Way, Suite 205, Tinton Falls, NJ	Never paid this entity	KHK6E1JW9C15 BOOZ ALLEN HAMILTON INC · Tinton Falls NJ. 1 of 41 children, the only one in NJ	Parent 0001443646 files as Booz Allen Hamilton Holding Corp; former name EXPLORER HOLDING CORP
S1 · THE PHONE	Call (703) 555-0147	Master number changed by email 3 days earlier	Not carried by USAspending	SEC number 703-902-5000. Demo scenario: do not call.
S2 · SAIC OR LEIDOS	Science Applications International Corporation	Two vendors fit: V-10442, V-10587	LEIDOS, INC. carries the alias SCIENCE APPLICATIONS INTERNATIONAL CORPORATION	Leidos Holdings, Inc. former name SAIC, Inc. (2005-2013)
S4 · V2X	Letterhead: Vectrus, Inc.	V2X on file	CG8SV4246V29 alias VECTRUS INC.; location null	0001601548 former names Vectrus, Inc. (2014-2022), Exelis MSCO Inc.

Every registry fact here is real public data. ■ DISAGREES ■ GAP OR INCONSISTENT ■ CONSISTENT

ONE REQUEST IN, ONE DECISION OUT

Solution

Not reconciling records: deciding whether to trust a claim before money moves.

✓ APPROVE | CALL BACK FIRST | × HOLD

A cited fact sheet. A call-back script that uses only independently sourced numbers. The exposure on the payment run. An unasked sweep of the rest of the run.



Code decides; models read, plan, critique and explain.

FOUR AGENTS, FOUR AUTHORITIES

Agent workflow and architecture

A1 · NEMOTRON 3 ULTRA

Letter Reader

Turns the letter into quoted claims

A2 · NEMOTRON 3.5 LIGHTNING

Registry Scout

A fixed floor of Zetaris checks, plus the checks the floor misses

A3 · NEMOTRON 3 ULTRA

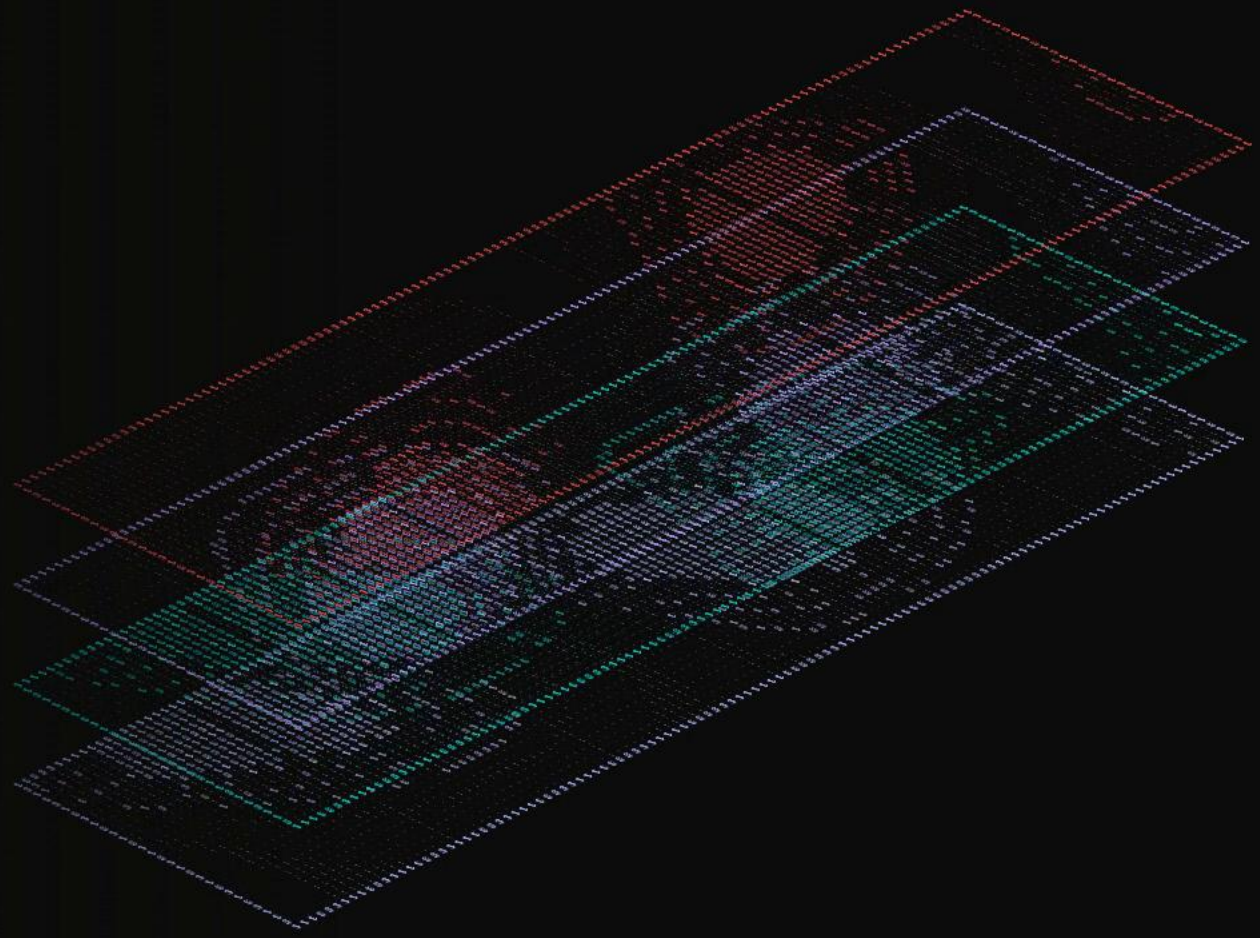
Identity Examiner

Block authority: re-queries, sends the Scout back, nobody overrides

A4 · CODE DECIDES

Gate Officer

Policy in code: approve, call back or hold; sweeps the run



GAPREQUEST LOOP, ≤2 ROUNDS

RE-QUERY · ROW HASH

FAIL CLOSED

WORLD MODEL: SHARED STATE

WHAT IT RUNS ON

Tech stack

LAYER 4

Interface

Next.js 15 · Tailwind v4 · Radix

LAYER 3

Agents and policy

TypeScript on Node 22 · Fastify · zod · pino

LAYER 2

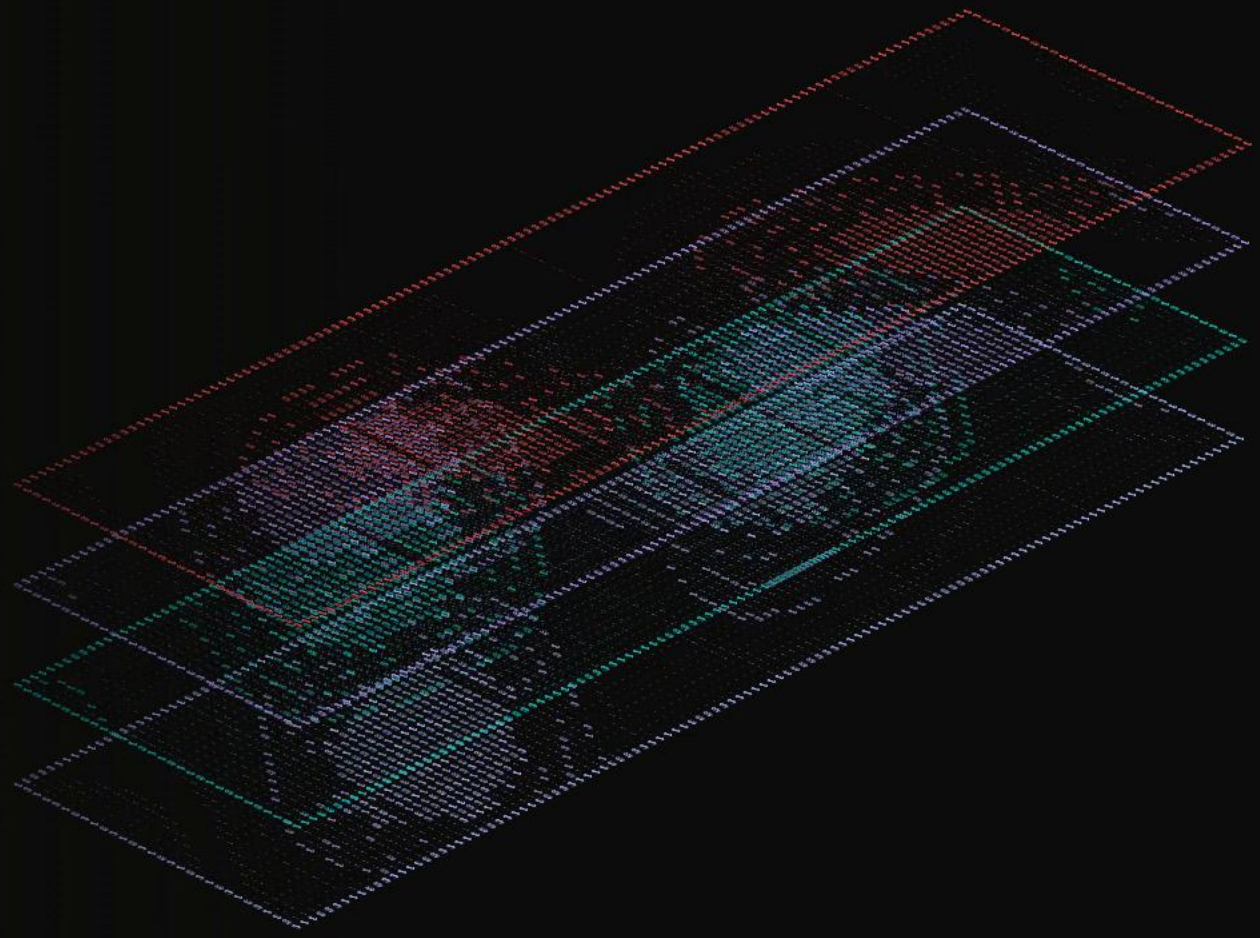
Evidence and memory

Zetaris (local) · Postgres 16 · Docker · Meterless World Model

LAYER 1

Models

Meterless Nightshift → NVIDIA Nemotron 3 Ultra and 3.5 Lightning



ZETARIS · UNIFIED SEMANTIC LAYER

How we used Zetaris

3 sources → one semantic layer, vendor360 (12 tables)

9 exploratory + 3 USL-active queries

DQ1 and DQ2 are the unasked sweep of the payment run

Every fact carries a query id you can re-run

- registry_entity
- registry_child
- registry_parent
- registry_alias
- sec_registrant
- vendor
- vendor_contact
- vendor_bank_account
- vendor_contact_log
- ap_invoice
- bank_change_request
- payment_run_item

Reflection docs: time saved, would it help, and a requirement we wish Zetaris had: keyed REST templates.



LIST DQ USL, RUN DQ



RE-RUN IN ZETARIS: MATCHES THE CITED ROWS

METERLESS · NVIDIA NEMOTRON

How we used Meterless and NVIDIA

WORLD MODEL

Bitemporal edges, point-in-time call-back, cross-session memory of a confirmed impersonation.

NIGHTSHIFT

Two pinned lanes: nvidia/nemotron-3-ultra-550b-a55b and nvidia/nemotron-3.5-lightning-30b-a3b. A per-call app id, the model logged per call, a receipt for each.

NEMOTRON'S FOUR JOBS, MEASURED QUALITY

Field accuracy and quote support, Ultra vs Lightning	AWAITING MEASURED F31
Injection caught n/4	AWAITING MEASURED F31
Examiner agreement n/6	AWAITING MEASURED F31

Examiner labels were written by our coding assistant (Claude Opus 5.5) from captured rows; not expert-labelled.

AWAITING RECORDED RUN



ENTITY GRAPH, A SUPERSEDED EDGE IN GREY



MODEL LEDGER: MODEL LOGGED PER CALL

MEASURED COST PER DECISION

AWAITING RECORDED RUN

FIVE THINGS WE DID DIFFERENTLY, THEN THE NUMBERS

What was unique, and results

The call-back number is chosen by provenance, not by convenience.

A Zetaris data-quality rule is the unasked action: the sweep of the whole run.

Discovery is narrowed by the claim: 41 Booz Allen entities, the one in the claimed state.

When the sources cannot say which vendor, it refuses to guess and writes a script for each.

Before a discovered link counts, the Examiner re-runs its decisive query and compares row hashes.



THREE PROCESSES, MEMORY CARRIED BY THE WORLD MODEL

MEDIAN DECISION TIME	ZETARIS QUERIES PER RUN	MODEL CALLS PER LANE	TESTS PASSED
AWAITING RUN	AWAITING RUN	AWAITING RUN	AWAITING RUN

HONEST EDGES

Limits and what comes next

- It does not prove who owns a bank account; no ownership validation service is connected.
- It never applies a change and never places a call. A human does both.
- Registry coverage at setup is 6 federal-contractor families; other entities are discovered at run time. US entities and USD only.
- SAM.gov entity data, Trade.gov CSL and OpenSanctions are not connected (keys required).
- Nemotron is claimed only for calls whose returned model matched; the model ledger shows which.

NEXT

- Bank-account ownership validation
- SAM.gov and exclusions as Zetaris sources
- Keyed REST templates
- Controller run view · ERP write-back behind approval

